

TORUN, Poland

WMO: 122500

Lat: 53.0422N

Lon: 18.5953E

Elev: 237

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.7	9.9	-0.9	5.3	4.9	4.4	7.0	11.0	17.6	40.5	15.9	37.5	3.1	60	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	86.9	67.5	83.3	66.0	80.1	64.5	70.0	82.1	68.2	78.9	66.6	76.3	7.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.3	73.8	64.5	91.8	72.5	62.9	86.6	70.9	34.0	82.1	32.6	79.0	31.3	76.1	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	13.1	11.7	DB	-2.1	92.9	9.2	2.8	-8.8	94.9	-14.2	96.5	-19.3	98.0	-26.0	100.0	(4)
			WB	-2.7	73.1	9.0	2.0	-9.2	74.5	-14.5	75.7	-19.5	76.8	-26.1	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	29.2	31.5	37.6	47.5	56.2	62.1	66.5	65.6	57.2	47.9	39.2	32.4	(6)
(7)		DBStd	15.10	9.87	9.36	7.87	7.61	7.53	6.39	5.91	5.67	6.11	6.98	6.93	8.85	(7)
(8)		HDD50	2714	646	518	389	134	24	0	0	8	124	325	546	(8)	
(9)		HDD65	6519	1111	937	850	525	291	130	53	62	246	530	773	1011	(9)
(10)		CDD50	1921	0	0	4	60	215	363	510	485	223	59	2	0	(10)
(11)		CDD65	252	0	0	0	1	17	43	98	82	11	0	0	0	(11)
(12)		CDH74	2629	0	0	0	40	219	495	988	776	111	0	0	0	(12)
(13)		CDH80	746	0	0	0	7	37	139	327	212	24	0	0	0	(13)
(14)	Wind	WSAvg	5.8	6.4	6.3	6.4	6.0	5.7	5.5	5.3	4.9	5.1	5.6	6.0	6.3	(14)
(15)	Precipitation	PrecAvg	21.2	1.1	0.9	1.1	1.3	2.1	2.8	3.2	2.5	1.9	1.5	1.5	1.5	(15)
(16)		PrecMax	33.3	2.2	2.1	2.0	4.1	4.9	11.8	9.3	6.3	6.1	5.1	3.2	3.1	(16)
(17)		PrecMin	12.3	0.1	0.1	0.2	0.2	0.6	0.8	0.7	0.1	0.3	0.2	0.4	0.3	(17)
(18)		PrecStd	4.3	0.6	0.5	0.5	0.8	1.0	1.9	2.0	1.2	1.3	1.0	0.6	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	52.2	63.0	79.2	84.3	90.6	92.4	90.2	83.3	70.4	57.0	52.1	(19)
MCWB			46.3	46.7	51.3	59.5	64.4	69.3	68.8	69.2	64.8	60.7	53.3	49.8	(20)	
2%		DB	46.3	49.1	57.8	72.3	79.8	83.7	87.4	85.2	76.8	65.7	53.6	48.5	(21)	
		MCWB	43.7	45.2	48.2	56.3	62.5	66.0	67.7	67.4	62.6	58.0	50.6	46.3	(22)	
5%		DB	43.4	46.2	53.2	66.8	75.8	79.5	83.6	81.7	72.4	61.8	51.0	45.9	(23)	
		MCWB	41.3	42.9	46.3	53.3	60.8	63.6	66.5	66.0	60.6	56.1	48.4	43.8	(24)	
10%		DB	40.9	43.3	49.1	62.0	71.2	75.7	79.5	78.3	68.6	58.6	48.7	43.3	(25)	
		MCWB	39.0	40.5	44.0	51.1	58.3	62.2	65.0	64.3	59.2	54.4	46.4	41.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	48.4	53.0	60.7	66.5	71.1	72.1	71.9	66.8	62.0	54.2	50.3	(27)
MCDWB			49.2	50.5	60.4	75.8	79.7	84.8	84.8	85.3	78.4	67.8	56.1	51.5	(28)	
2%		WB	44.0	45.6	49.7	57.6	64.2	68.3	70.2	69.4	64.5	59.0	51.0	46.7	(29)	
		MCDWB	46.0	48.6	55.7	69.7	76.4	79.7	82.0	81.2	73.9	63.6	53.1	48.2	(30)	
5%		WB	41.5	43.1	46.9	54.7	62.0	65.9	68.5	67.8	62.2	56.7	48.6	43.8	(31)	
		MCDWB	43.3	46.0	52.0	64.5	72.8	76.3	79.1	77.8	69.3	61.0	50.6	45.5	(32)	
10%		WB	39.1	40.6	44.5	52.1	59.8	63.6	66.8	66.1	60.3	54.8	46.6	41.7	(33)	
		MCDWB	40.8	43.1	48.7	60.2	69.2	72.6	76.1	75.0	66.9	58.4	48.6	43.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	10.9	14.2	19.4	20.3	19.5	18.9	19.6	17.3	13.9	8.7	7.8	(35)
MCDBR			8.6	13.4	21.0	28.0	26.2	26.1	25.5	25.3	23.6	18.6	11.5	9.2	(36)	
MCWBR			7.6	10.2	13.3	14.3	11.9	11.4	9.6	10.6	11.5	11.8	9.1	8.0	(37)	
5% WB		MCDWB	8.6	12.5	18.5	23.9	22.9	22.1	21.4	21.5	19.8	16.8	11.1	8.9	(38)	
		MCWBR	7.8	10.0	12.6	12.9	11.3	11.4	9.6	10.1	10.8	11.5	9.3	8.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.340	0.373	0.420	0.414	0.407	0.441	0.429	0.386	0.365	0.347	0.309	(40)	
(41)		taud	2.257	2.264	2.239	2.163	2.236	2.286	2.210	2.246	2.341	2.382	2.325	2.287	(41)	
(42)		Ebn at Noon	207	238	254	256	265	268	255	251	249	228	192	185	(42)	
(43)		Edh at Noon	20	27	34	42	41	40	42	38	31	24	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	200	420	814	1319	1659	1730	1587	1397	1002	545	221	143	(44)	
(45)		RadStd	36	66	110	171	168	159	186	126	131	82	29	18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (29 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	-299	+144	N/A	(47)

Nomenclature: See separate page